

Explainable Semantic Prototype-Augmented Ordinal Contrastive Learning for Imbalanced Medical Severity Grading

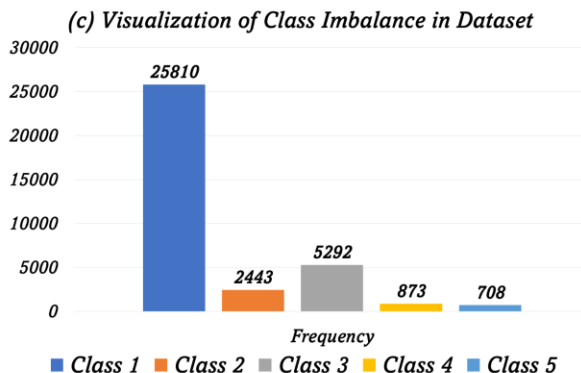
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Core problem: Medical severity grading models predict labels
but cannot explain *why*

Problem & Goal

The Setting

- Ordinal prediction: labels carry ordering (Grade 0 → 4)
- Datasets: BUSI (780 images, 3 classes) and DR (35,126 images, 5 grades)
- Severe class imbalance, DR: 74% No-DR, 2% PDR



Core Limitation

- Standard models optimize cross-entropy with no ordinal awareness
- No connection to clinical definitions underpinning each grade
- Cannot identify *which* clinical features drove a prediction

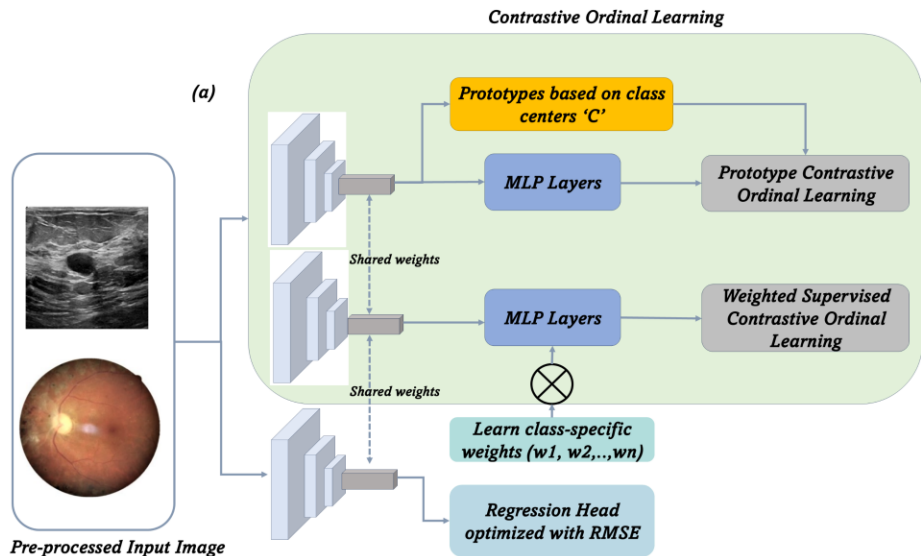
Goal: add semantic + spatially grounded explainability to ordinal contrastive learning

Baseline

A Hybrid Contrastive Ordinal Regression Method for Advancing Disease Severity Assessment in Imbalanced Medical Datasets

Architecture details

- EfficientNet-V2S backbone extracts 1280-dimensional image features
- PCOL: prototype-based contrastive learning that aligns samples with class prototypes while preserving ordinal severity relationships
- SCOLw: weighted supervised contrastive learning that improves minority-class representation using inverse-frequency class weighting
- Regression head predicts a continuous severity score used for final disease grading

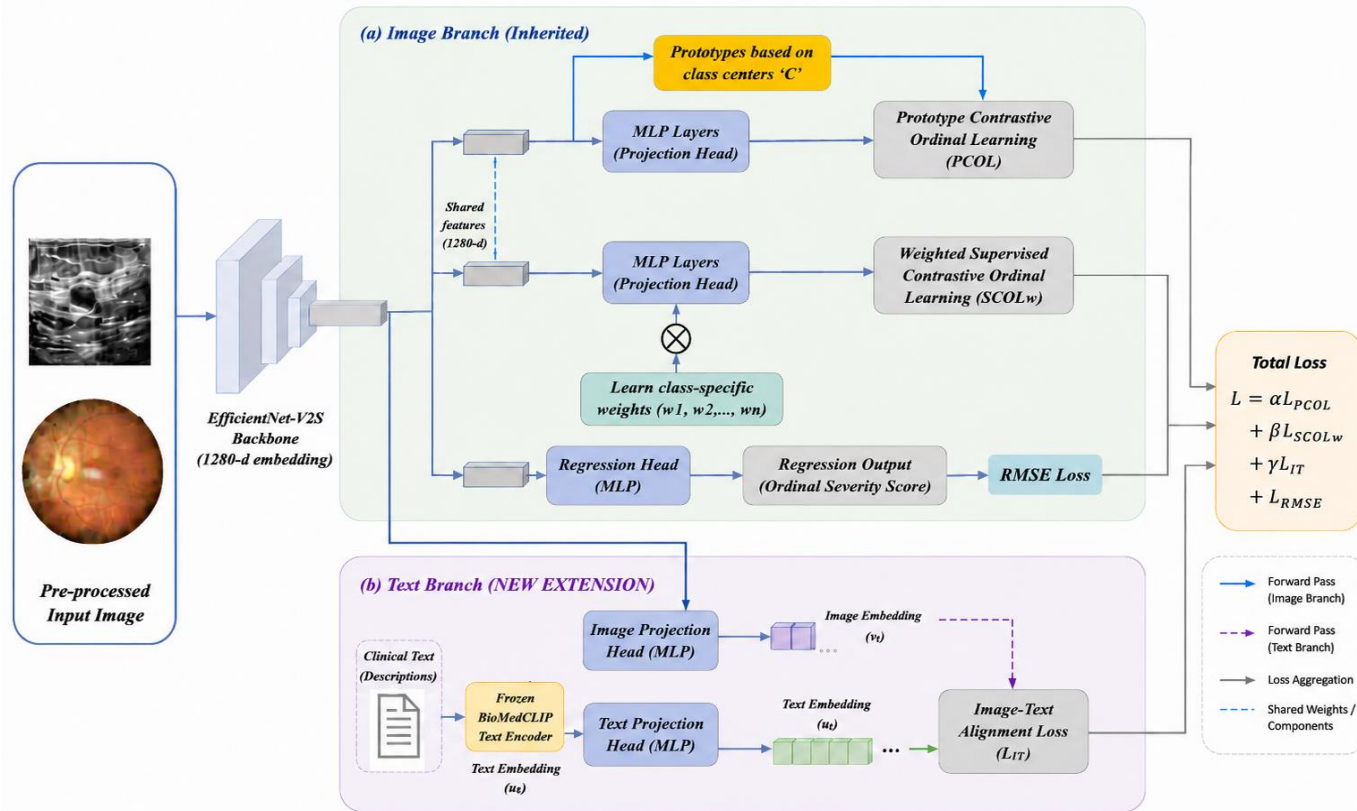


$$L_{total} = \alpha L_{PCOL} + \beta L_{SCOLw} + L_{RMSE}$$

$$L_{total} = \alpha \left(- \sum_{i \in I} \log \frac{\exp(f_a^\top c_p / \tau)}{\sum_{c_n \in N(i)} \exp((f_a^\top c_n + r_{a,n}) / \tau)} \right) +$$

$$\beta \left(\sum_{i \in B} \frac{-w_i}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(f_a^\top f_p / \tau)}{\sum_{n \in N(i)} \exp((f_a^\top f_n + r_{a,n}) / \tau)} \right) + \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2}$$

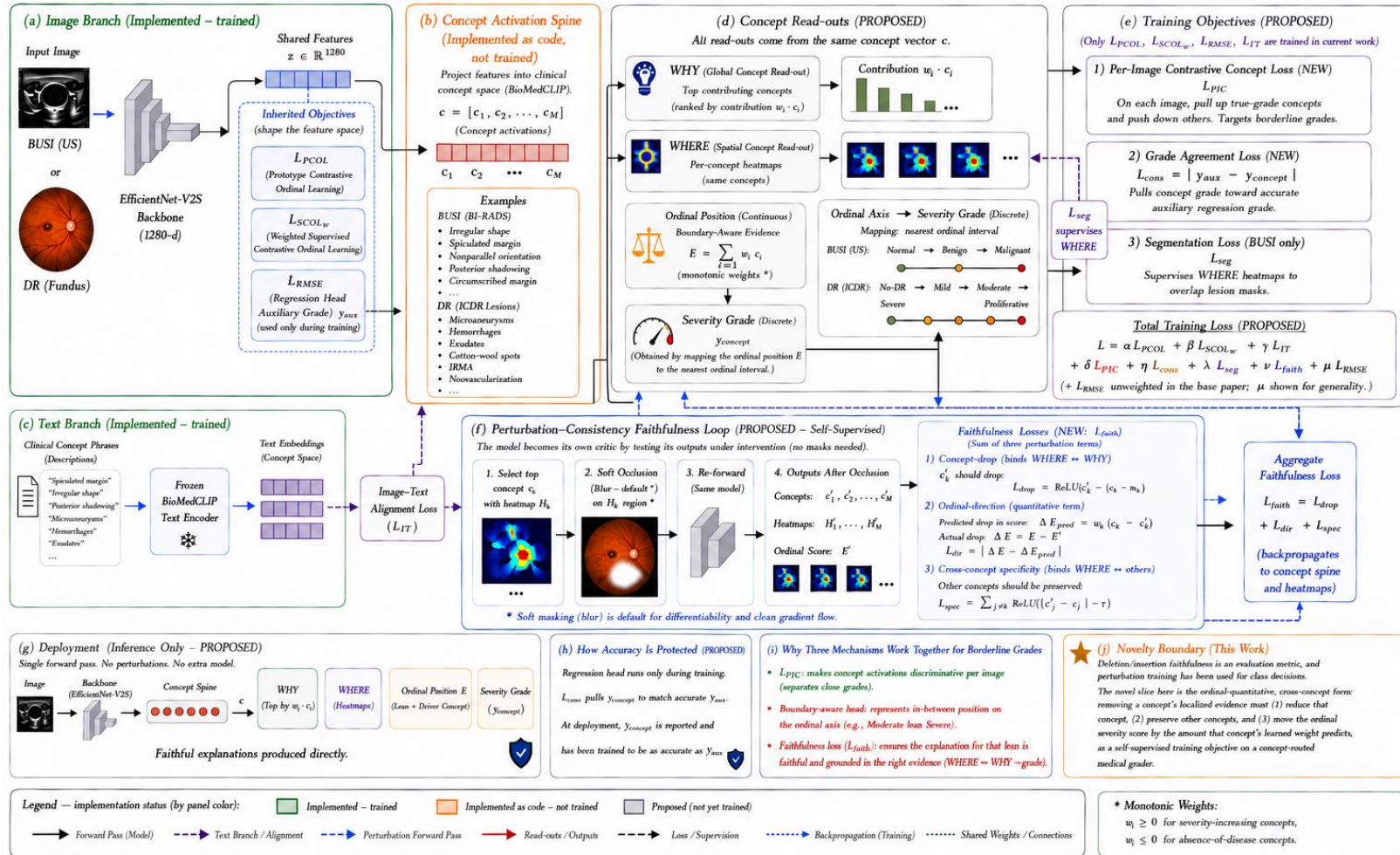
Hybrid Contrastive Ordinal Learning with Semantic Extension



$$L_{IT} = - \sum_{i \in I} \log \frac{\exp(\text{sim}(v_i, u_{y_i})/\tau)}{\sum_{k=1}^K \exp((\text{sim}(v_i, u_k) + \lambda|y_i - k|)/\tau)}$$

ESPAOCL: Explainable Self-supervised Perturbation-Consistency Ordinal Concept Learning (PROPOSED)

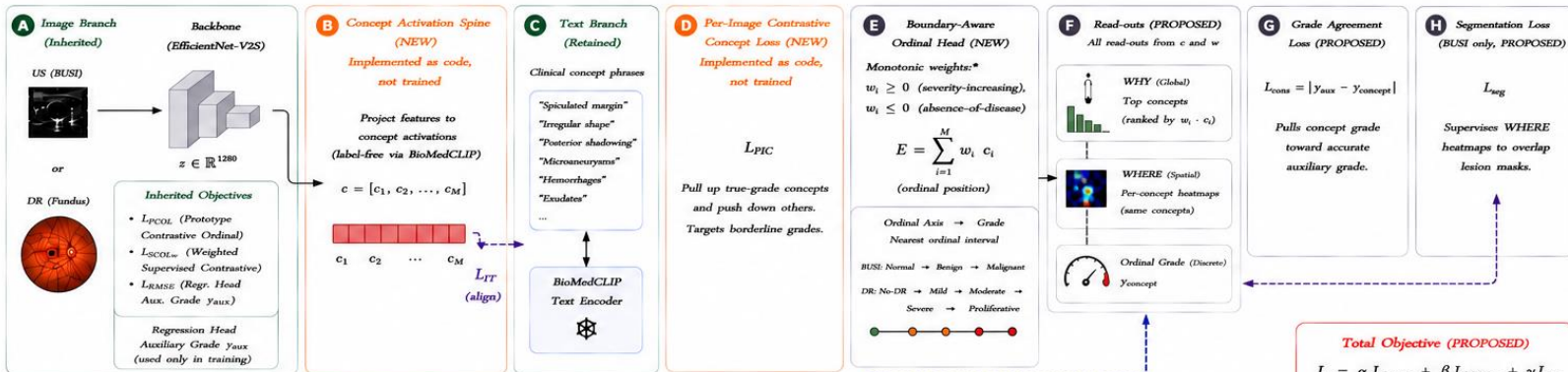
STATUS: Only the inherited image and text branches are trained; the concept-routing architecture is proposed, with the spine and L_{PIC} coded but untrained. No external API calls anywhere.



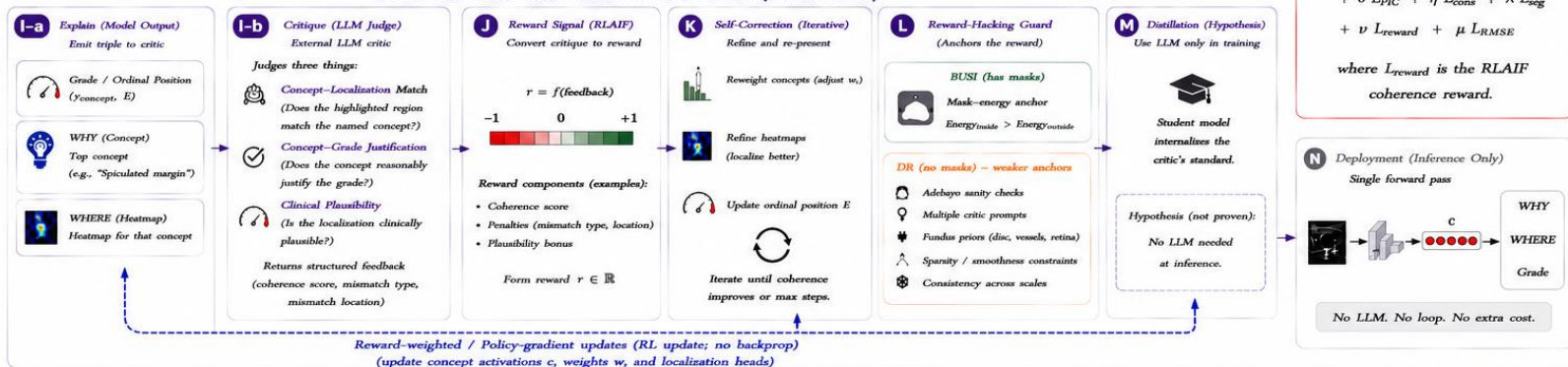
Direction 1: LLM-in-the-loop Critic (Paid, External-Judge Version) (Training-time only; distilled out at inference)

One Architecture, Two Faithfulness Paths: A-H are shared. Faithfulness is enforced either by 1 LLM critic (this diagram) or by 2 Self-supervised perturbation (Direction 2).

SHARED BASE (A-H)



LLM-IN-THE-LOOP FAITHFULNESS MECHANISM (TRAINING ONLY)



Legend (status)

- Implemented & trained
- Implemented as code, not trained
- Proposed (not yet trained)

Legend (flows)

- Forward pass (model)
- Loss / Supervision
- RL update (no backprop)
- Read-outs / Outputs
- Shared weights / connections

★ Novelty (Narrow Claim)

RLAIF and LLM-as-judge exist for generative text.
The novel Alice is using an LLM coherence critic as a training signal to align a discriminative medical grader's concept explanations with their localizations, in a closed loop, and distilled out at inference.

* Monotonic weights:

- $w_1 \geq 0$ for severity-increasing concepts;
- $w_1 \leq 0$ for absence-of-disease concepts.

Replication Results

Paper targets vs. our first reproduction and New reproduction

Dataset	Paper	Our Old	Our New
BUSI (5-fold CV)	91.0% acc / 0.10 MAE	89.0% $\pm$ 1.5% acc / 0.150 MAE	91.53% acc/ 0.0911 MAE
DR (10-fold CV)	83.8% acc / 0.21 MAE	74.6% acc / 0.350 MAE	82.03% acc / 0.2478 MAE

- Initial replication did not match the paper results on BUSI and DR.
- We re-implemented the framework in a modular codebase and verified each component individually.
- Applying implementation fixes, author clarifications, and hyperparameter tuning significantly improved performance and brought the results closer to the paper.

DR Replication: From 74.6% to 82.0%

Key implementation corrections and tuning decisions that closed the gap to the paper

Implementation Changes

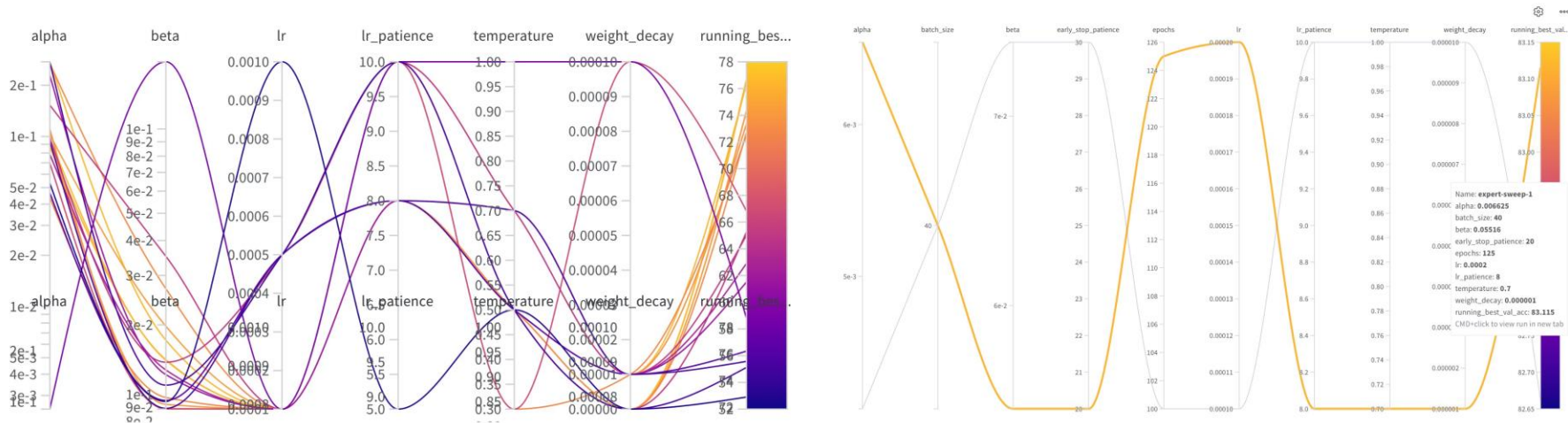
- Corrected SCOLw class weighting using batch-level inverse frequencies, following the authors' rebuttal clarification.
- Updated PCOL and SCOLw ordinal penalties to use raw label distances $|y_i - y_j|$ instead of normalized distances.
- Added ImageNet mean/std normalization for EfficientNet-V2S pretrained weights.
- Introduced BatchNorm in projection heads for more stable contrastive representation learning.

Training Improvements

- Tuned loss weights controlling contrastive learning:
 - $\alpha = 0.00662$
 - $\beta = 0.05516$
- Tuned contrastive temperature:
 - $\tau = 0.7$
- Reduced learning rate:
 - $1e-3 \rightarrow 2e-4$
- Increased early stopping patience and scheduler patience to improve convergence stability.

DR Replication: From 74.6% to 82.0%

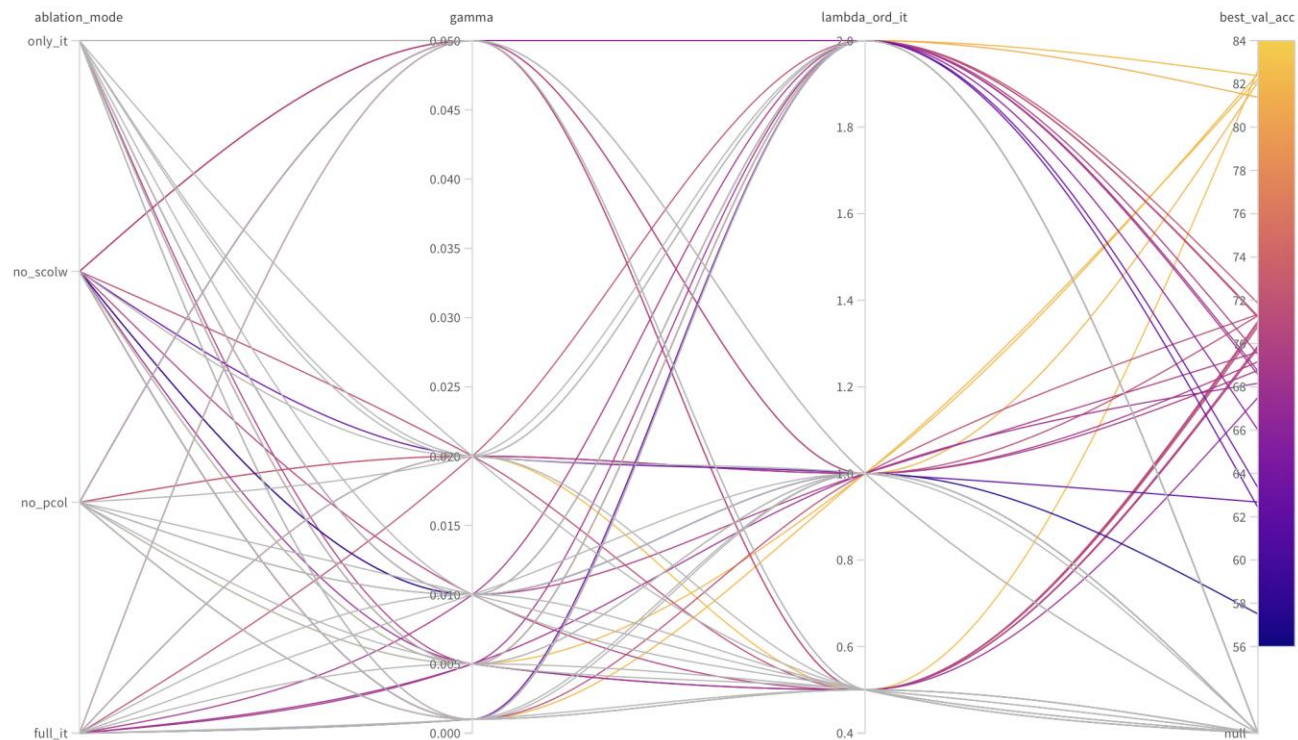
Most important tool was wandb sweep



Weight and Bias Bayesian Optimization: Instead of manually tuning parameters, the sweep iteratively searched for better configurations based on prior results. This was one of the key tools used to close the gap to the paper's reported performance.

Ablation Experiment on our New Architecture

Evaluating the contribution of the image-text ordinal alignment module through ablation experiments that vary γ and λ while selectively removing PCOL and SCOLw to quantify the effect of each component on grading accuracy and MAE.



The γ Problem

Why the LIT Loss Broke DR Accuracy — and How to Fix It

We added $\gamma = 0.1$ and DR accuracy dropped 11.9 pp. This proposal explains why — and proposes a principled fix.

The Full Training Loss

Four losses run simultaneously — each pulling the model in a different direction.

$$L = \alpha \cdot L_{\text{PCOL}} + \beta \cdot L_{\text{SCOLw}} + \gamma \cdot L_{\text{IT}} + L_{\text{RMSE}} \quad (\text{Eq. 9})$$

$$\alpha = 0.01 \quad \beta = 0.10 \quad (0.001 \text{ on DR}) \quad \gamma = 0.1 \leftarrow \text{focus} \quad L_{\text{RMSE}} \text{ weight} = 1.0$$

- | | | |
|--|--|--|
| ● L_{PCOL} | <i>Prototype contrastive</i> | Groups same-grade images together, pushes different grades apart. Grades further apart get pushed harder. |
| ● L_{SCOLw} | <i>Weighted sample contrastive</i> | Same as PCOL but image-to-image. Rare grades (PDR: 2%) get higher weight to avoid being ignored by majority class. |
| ● $\gamma \cdot L_{\text{IT}}$ | <i>Image-text alignment</i> $\leftarrow$ THIS IS THE PROBLEM | Pulls each image toward its grade's clinical text description (BioMedCLIP). Controlled by γ . Our focus. |
| ● L_{RMSE} | <i>Regression — the primary task, weight 1.0</i> | Predicts the severity grade 0–4. This is what accuracy measures. The most important loss. |

What We Did: The $\gamma = 0.1$ Experiment

BUSI — 780 IMAGES · 3 CLASSES

Without LIT ($\gamma = 0$)

87.8%

accuracy

With LIT ($\gamma = 0.1$)

85.9% **-1.9 pp**

Small drop — acceptable trade-off.

DR — 35,126 IMAGES · 5 GRADES · 74% NO-DR

Without LIT ($\gamma = 0$)

73.3%

accuracy

With LIT ($\gamma = 0.1$)

61.4% **-11.9 pp**

Primary task collapsed. Not acceptable.

why did BUSI survive $\gamma = 0.1$ but DR dropped by 11.9 pp... The γ ablation was only run on BUSI

Open Question Regarding γ

γ is a fixed scalar

γ was chosen by trying four values (0, 0.1, 0.2, 0.5) on BUSI, picking the best concept margin, and keeping it fixed throughout training on both datasets. That is a hyperparameter, not a learnable weight.

Why DR Collapsed — Three possible reasons

REASON 1 — LOSS SCALE MISMATCH

```
L_RMSE ≈ 0.40      L_IT ≈ 1.20      →      γ · L_IT = 0.1 × 1.20 = 0.12
Real share of L_IT = 0.12 / (0.40 + 0.12) = 23%, not 10% as intended.
And this ratio changes every epoch — a fixed γ does not mean a fixed contribution.
```

REASON 2 — CONFLICTING OBJECTIVES ON DR

L_RMSE says: focus on No-DR — 74% of DR images are grade 0, that is where accuracy comes from. **L_IT says:** treat all 5 grades equally regardless of frequency. **These are opposite instructions. The backbone cannot fully satisfy both — accuracy drops.**

REASON 3 — THE γ ABLATION WAS NEVER RUN ON DR

Table 7 in the paper shows $\gamma \in \{0, 0.1, 0.2, 0.5\}$ results for BUSI only. For DR, only $\gamma = 0.1$ is reported (Table 6). **The parameter that caused the 11.9 pp drop was never tested on DR before being applied.**

Why did BUSI survive? Fewer classes (3 vs 5), less imbalance, and smaller L_IT values by design.

The Fix — Adaptive Loss Ratio Normalisation

CURRENT — BROKEN

```
 $\gamma = 0.1$ 
```

```
Fixed. Never changes.  
23% of total on DR,  
not the 10% intended.
```

Picked by trying 4 values on BUSI. No mathematical justification. Means something different every epoch.

PROPOSED — FIX

```
 $\gamma_t = f \times (L_{\text{RMSE}} / L_{\text{IT}})$ 
```

```
f = 0.1 (start here)  
Recalculated every step.  
 $\gamma_t \cdot L_{\text{IT}} = f \times L_{\text{RMSE}}$ . Always.
```

f has a clear meaning: fraction of RMSE's signal given to text alignment. Guaranteed constant contribution on every dataset.

Is This Feasible? — Honest Check

WHY IT WILL WORK

- One new hyperparameter f , starting at 0.1. Sweep is $\{0.05, 0.1, 0.2\}$ — same cost as the original γ ablation, this time on DR.
- Runs immediately on new compute setup. ~2 hrs per fold.

RISKS TO WATCH

- **Gradient direction conflict may remain.**

If L_{IT} and L_{RMSE} pull weights in opposite directions (not just different scales), ratio normalisation fixes scale but not direction. Diagnostic: log gradient norms. If $||\nabla L_{IT}|| \gg ||\nabla L_{RMSE}||$ after the fix, GradNorm is the next step.

- **$f = 0.1$ may still be too high for DR.**

Handled by Experiment E3 — sweep $f = 0.05$ and $f = 0.2$. Cheap and identifies the right fraction directly.

Required Experiments

Five experiments in order of priority. E1 and E4 are free. E2 is the critical test.

E1 **γ ablation on DR**

Run $\gamma = 0.2$ and 0.5 on DR — paper only ever tested γ on BUSI. Confirms whether accuracy drops further as γ increases.

E2 **Dynamic γ_t with $f = 0.1$ on DR ← KEY**

Does keeping L_{IT} at a constant 10% of L_{RMSE} prevent the 11.9 pp accuracy collapse? Run 3 folds first, then full 10 if confirmed.

E3 **f sweep: 0.05 and 0.2 on DR**

Find the fraction f that maximises Accuracy + $0.5 \times$ Concept Margin jointly. Prevents repeating the original mistake of optimising one metric only.

E4 **Log L_{RMSE} and L_{IT} separately every epoch**

Directly verifies the 23% scale mismatch calculation. If the ratio $L_{IT} / L_{RMSE} > 1$ at $\gamma = 0.1$ on DR, the diagnosis is confirmed.

E5 **Apply best dynamic γ_t to BUSI as sanity check**

BUSI was not broken. Confirm the fix keeps BUSI accuracy $\geq 85.9\%$ and concept margin ≥ 1.05 .

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The resolution-collapse problem

Why plain Grad-CAM on the last layer can't see what we need

EfficientNet-V2S compresses too aggressively

- **Input image**
 - 300×300 pixels
- **Final feature map**
 - 7×7 cells
 - Each cell covers $\lceil 300/7 \rceil \approx 43$ pixels
- **A microaneurysm is 2–4 pixels wide**
 - Sub-cellular at the final layer
 - Plain Grad-CAM dilutes the lesion signal by $\approx (43/3)^2 \approx 200\times$

Our fix: hook four stages at once

- Stage 2 $\rightarrow 75 \times 75$ (≈ 4 px/cell)
- Stage 4 $\rightarrow 19 \times 19$ (≈ 16 px/cell)
- Stage 6 $\rightarrow 10 \times 10$ (≈ 30 px/cell)
- Stage 7 $\rightarrow 7 \times 7$ (≈ 43 px/cell)
- **Then upsample everything to 300×300 and pixel-wise MAX-fuse**

Fine detail from early layers, semantic context from late layers, in one heatmap

Grad-CAM++ on each hooked layer

Second-order gradient weighting picks lesion peaks instead of averaging them

Per-pixel weight (Grad-CAM++)

$$\alpha(i,j,k) = (G(i,j,k))^2 / [2 \cdot (G(i,j,k))^2 + \sum_k A(k) \cdot (G(k))^3 + \epsilon]$$

Why second-order?

- First-order Grad-CAM averages all gradients equally
- α is large where the gradient peaks (lesion centres)
- α is small in flat background
- *Substantially better contrast on small lesions*

Channel importance

$$w(k) = \sum \alpha(i,j,k) \cdot \text{ReLU}(G(i,j,k))$$

- Each channel gets a weight by aggregating its peaks
- ReLU keeps only positive evidence (no dropouts)

Per-layer attention map

$$\text{CAM}(\ell) = \text{ReLU}(\sum_k w(k) \cdot A(k))$$

- Weighted sum of feature channels
- ReLU drops negative contributions

Multi-scale fusion: MAX across all four layers

Pixel-wise max preserves the sharpest signal from whichever layer detected each feature

$$\text{CAM_final}(i,j) = \max_{\{\ell \in \{2,4,6,7\}\}} \text{Up}(\text{CAM}(\ell), 300)$$

Why MAX, not mean

- MAX picks whichever layer fires hardest at each pixel
- Stage 2 wins on tiny features (microaneurysms)
- Stage 7 wins on larger semantic features
- *Mean would smooth them all together and lose the small ones*

What this gets us

- Microaneurysm-scale spatial detail in DR images
- Lesion-scale localization on BUSI ultrasound
- All four layer hooks are computed in a single forward+backward pass
- *Post-hoc: runs on any saved checkpoint, ~15 min per dataset on T4*

Concept-guided LayerCAM: the WHY substitution

Replace the gradient target. One line of code, very different question.

Standard LayerCAM

$$s = - | \hat{y} - y_{\text{target}} |$$

Asks: which pixels drove Grade 2?

- Heatmap is class-discriminative
- Same map regardless of which clinical feature we ask about
- *Cannot answer concept-level questions*

Concept-guided LayerCAM (ours)

$$s = \hat{h}_{\text{image}} \cdot v_{\text{concept}}$$

Asks: which pixels look like spiculated margins?

- One backward pass per concept phrase
- Produces a 10-panel (BUSI) or 11-panel (DR) grid
- Each panel is a spatially-localized answer to a different clinical question
- *LIME and SHAP can't do this — they have no differentiable gradient target*

WHY: concept similarity scoring

Cosine similarity between image embedding and aligned concept vector

$$a_j = \text{sim}(\hat{z}_i, \hat{v}_j) = (z_i \cdot v_j) / (\|z_i\| \cdot \|v_j\|) \in [-1, +1]$$

Score near +1

- Image strongly resembles this clinical feature

Example: Severe NPDR image scoring +0.88 on "venous beading"

Score near 0

- Image neither resembles nor contradicts the feature

Common at boundary cases between two grades

Score near -1

- Image is dissimilar from this feature

Example: PDR image at -0.087 on "normal retina"

Concepts are distinct and class-separated

Mean cosine similarity per (image-class, concept-group), fold 1, $\gamma = 0.1$

Each row is a test image class. Each column is a clinical-concept group. Bold cells = highest score per row (the diagonal).

Test class →	Normal concepts (no mass, no dist.)	Benign concepts (oval, circumscribed)	Malignant concepts (spiculated, irreg., ...)
Normal images (n=27)	+0.940	-0.201	-0.251
Benign images (n=87)	-0.191	+0.928	+0.034
Malignant images (n=42)	-0.252	+0.060	+0.946

The good news, and the catch

- Diagonal dominance: each test class scores its own concepts roughly 1.1 to 1.2 above the off-diagonal — clean separation
- **But within a class, all concepts of that class score nearly identically — heatmaps inside a class are visually interchangeable**

Which concepts fire, averaged over all 10 folds

Mean $\pm$ std across folds — anchored grades rank ~ 2.5 , relational grades sink to ~ 3.8

DR concept	grade	top-1 $\pm$ std	rank $\pm$ std
normal retina	No DR	0.42 $\pm$ 0.16	2.4 $\pm$ 0.1
healthy fundus	No DR	0.18 $\pm$ 0.17	2.5 $\pm$ 0.2
microaneurysms	Mild	0.38 $\pm$ 0.02	2.7 $\pm$ 0.0
dot/blot hemorrhage	Mod	0.19 $\pm$ 0.04	3.9 $\pm$ 0.3
hard exudates	Mod	0.13 $\pm$ 0.06	3.9 $\pm$ 0.3
cotton wool spots	Mod	0.08 $\pm$ 0.07	3.9 $\pm$ 0.3
venous beading	Sev	0.30 $\pm$ 0.08	2.8 $\pm$ 0.2
IRMA	Sev	0.37 $\pm$ 0.06	2.8 $\pm$ 0.1
neovascularization	PDR	0.33 $\pm$ 0.17	3.7 $\pm$ 0.5
preretinal hemorrhage	PDR	0.13 $\pm$ 0.16	3.7 $\pm$ 0.6
vitreous hemorrhage	PDR	0.19 $\pm$ 0.18	3.7 $\pm$ 0.5

Read the rank

- Anchored grades (No DR, Mild, Severe): rank ~ 2.4 – 2.8 .
- Moderate concepts: rank ~ 3.9 — buried, every fold.
- *std is tiny on ranks (0.04–0.56): the gap is consistent, not luck.*
- PDR top-1 std is large (few images, $n \approx 76$).

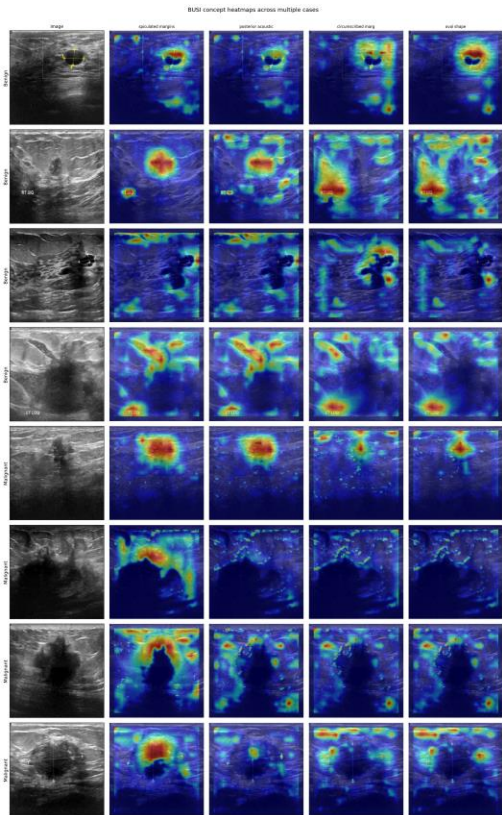
BUSI concepts over all 5 folds: every concept contributes

Single-fold zeros were unrepresentative — averaged, all concepts win some of the time

BUSI concept	class	top-1 $\pm$ std	rank $\pm$ std	Why this matters - On a single fold, 3 concepts read 0.00 — an artifact of that split. - Across 5 folds none is zero (0.18–0.53): all concepts carry signal. - Ranks cluster tightly (~1.8–3.0): concepts compete closely within a class.
no mass	Normal	0.53 $\pm$ 0.22	1.8 $\pm$ 0.3	
no distortion	Normal	0.42 $\pm$ 0.25	1.9 $\pm$ 0.4	
oval shape	Benign	0.22 $\pm$ 0.10	2.8 $\pm$ 0.3	
circumscribed margin	Benign	0.27 $\pm$ 0.10	3.0 $\pm$ 0.5	
parallel orientation	Benign	0.20 $\pm$ 0.15	2.9 $\pm$ 0.1	
posterior enhancement	Benign	0.22 $\pm$ 0.11	2.8 $\pm$ 0.2	
irregular shape	Malig	0.34 $\pm$ 0.14	2.9 $\pm$ 0.3	
spiculated margins	Malig	0.22 $\pm$ 0.13	2.8 $\pm$ 0.1	
nonparallel orientation	Malig	0.20 $\pm$ 0.13	2.9 $\pm$ 0.1	
posterior acoustic shadow	Malig	0.18 $\pm$ 0.17	2.6 $\pm$ 0.3	

Concept heatmaps across many cases, not one

Each concept asks WHERE its feature appears — distinct maps, still diffuse (the WHERE gap)



How to read each row

- Column 1 is the ultrasound image; the other four are heatmaps, one per clinical concept.
- Red = where the model sees that feature. A good explanation puts the right concept's heat on the lesion.
- The maps differ between concepts (the identical-heatmap bug is fixed) but still spread beyond the lesion.
- *Dark lesions read blue: the model keys on the bright rim, so heat forms rings around the mass, not on it.*
- **This diffuseness is exactly what the segmentation head and the auxiliary-seg loss target.**

Localization across the full fold, not single cases

Three complementary metrics separate location, boundary, and peak

BUSI	pointing	energy raw	energy guid	seg Dice
Benign	0.24	0.12	0.69	0.70
Malignant	0.62	0.29	0.69	0.52

What each metric isolates

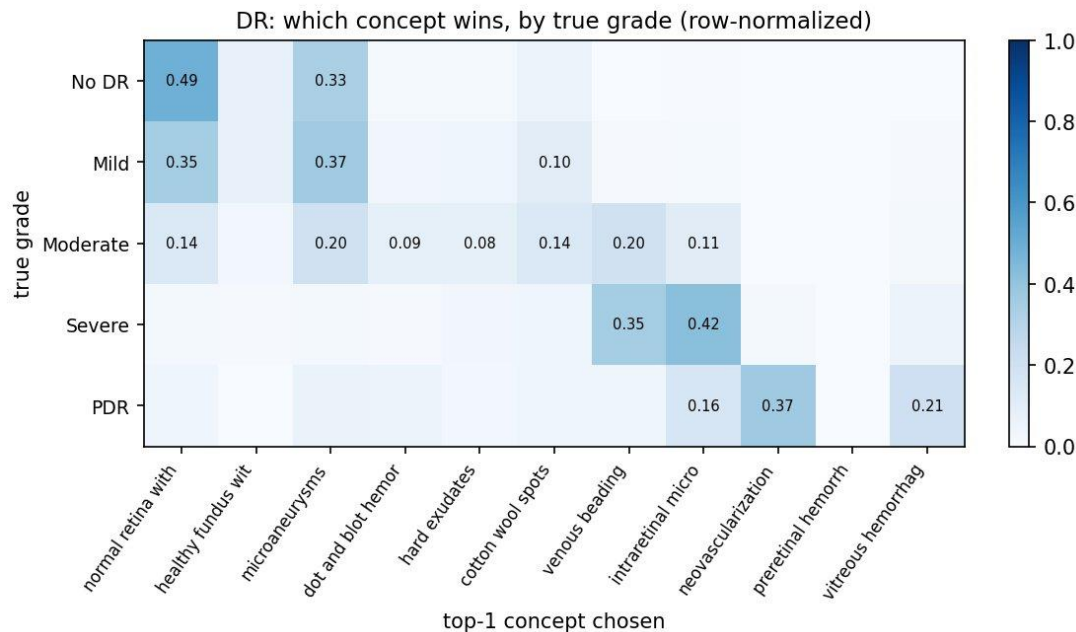
- pointing = is the single hottest pixel on the lesion (peak).
- energy = fraction of heatmap mass on the lesion (spread). Raw is diffuse; seg-guided concentrates it.
- seg Dice = boundary quality of the predicted lesion mask.

The honest reading

- Location is ~70% both classes (COM inside).
- Benign boundary is cleaner (Dice 0.70) than Malignant (0.52, spiculated).
- *energy-guided is segmentation-guided — report it as such, raw stays 0.12 / 0.29.*

Why Moderate scatters: the concept-confusion matrix

Each row is a true grade; color shows which concept won. Anchored grades concentrate; Moderate sprays



Reading the rows

- No DR and Severe: one or two concepts dominate the row — a clear winner.
- Moderate: the row sprays across seven concepts (0.08–0.20), no winner.
- This is the bimodal-scatter made visible: Moderate images land near both healthy and severe concepts.

What does not work yet: the Moderate grade

Two reinforcing reasons — the problem is where the images sit, not where the concepts sit

Reason 1 — no signature lesion

- ICDR defines Moderate as “more than Mild, less than Severe.”
- Every other grade has a defining lesion (microaneurysms, venous beading, neovascularization).
- *Moderate is defined by exclusion — nothing distinctive to match a concept against.*

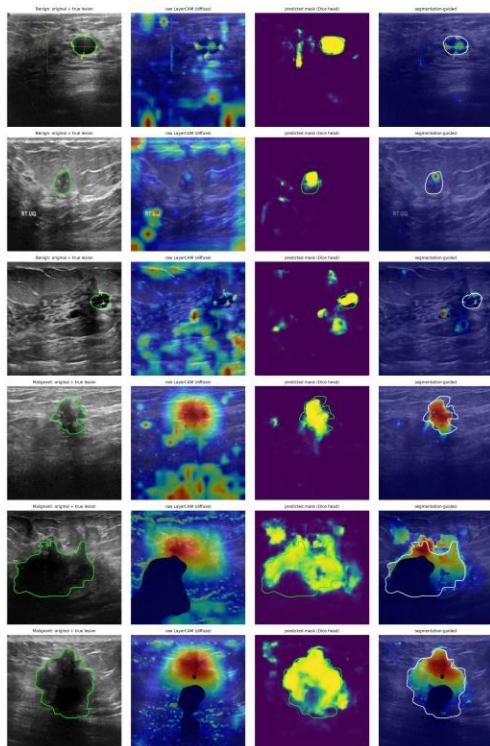
Reason 2 — bimodal embeddings

- Moderate images do not cluster: ~6/10 sit near Severe, the rest near No-DR.
- Their nearest concept is usually a neighbor-grade concept.
- Three post-hoc strategies (single / multi-prototype / adaptive) all left Moderate at chance.

Pins vs dots: post-hoc methods move the concept pins, but the image dots are scattered. You must retrain to move the dots.

What does not work yet: raw attention is diffuse

Four columns: image + true lesion · raw LayerCAM · predicted mask · segmentation-guided



What the columns show

- Col 2 (raw LayerCAM): heat is scattered across the whole image — only 12–29% lands on the lesion.
- Col 3 (predicted mask): the segmentation head finds the lesion region (Dice 0.52–0.70).
- Col 4 (guided): focusing the heat through the mask concentrates it on the lesion — energy rises to 0.69.

Honest caveat

- *The mask does the focusing; the raw attention is still diffuse. The deeper fix is training the seg loss into the backbone.*

Proposed loss 1: per-image contrastive concept loss

Forces the correct-grade concept to win on each individual image — the fix for Moderate

$$\mathcal{L}_{pic} = -\frac{1}{N} \sum_i \log \frac{\sum_{c \in G(y_i)} \exp(\text{sim}(z_i, v_c)/\tau)}{\sum_c \exp((\text{sim}(z_i, v_c) + \lambda |y_i - g(c)|)/\tau)}$$

z_i = image embedding v_c = concept vector $G(y_i)$ = concepts of image i 's true grade $g(c)$ = grade of concept c τ = temperature λ = ordinal penalty

Numerator

- Sum over the image's OWN-grade concepts.
- Rewards the correct-grade concepts scoring high on this image.

Denominator

- Sum over ALL concepts, with an ordinal penalty on wrong-grade ones.
- *A Moderate image picking Severe is penalized less than picking No-DR.*

What it does

- Per-image supremacy: the right concept must win on each image, not just on the class average.
- Reshapes the space so Moderate images stop landing in neighbor territory.

Proposed loss 2: auxiliary segmentation loss

Trains the backbone to concentrate attention on the lesion — sharpens RAW heatmaps (BUSI)

$$\mathcal{L}_{seg} = BCE(\sigma(\hat{s}), M) + \left[1 - \frac{2 \sum \sigma(\hat{s}) M + 1}{\sum \sigma(\hat{s}) + \sum M + 1} \right]$$

What each term is

- $\hat{s}$ = segmentation logits from a small decoder on the backbone feature maps.
- M = ground-truth lesion mask (BUSI ships these).
- BCE term: per-pixel correctness. Dice term: overlap, robust to the small-lesion class imbalance.

Why it sharpens (not just masks)

- The seg gradient flows INTO the backbone, so the features themselves learn to concentrate on the lesion.
- Frozen-head Dice (0.52–0.70) already proves the signal is present.
- *BUSI only; DR has no masks (use IDRiD to evaluate).*

Sharpening localization with a segmentation head

Raw heatmaps are diffuse; a frozen-backbone seg head focuses them onto the lesion

BUSI class	seg Dice	energy raw	energy guided
Benign	0.70	0.12	0.69
Malignant	0.52	0.29	0.69

Honest caveat

This is a segmentation-GUIDED heatmap. The mask does the focusing; the raw attention is still diffuse. Report both numbers.

What the numbers mean

- Energy-in-lesion = fraction of heatmap mass inside the lesion mask. Raw attention puts only 12–29% on the lesion; the seg head raises this to ~69%.

The combined training objective

Six terms: the four existing losses plus the two new ones

$$\mathcal{L} = \alpha \mathcal{L}_{PCOL} + \beta \mathcal{L}_{SCOLw} + \gamma \mathcal{L}_{IT} + \delta \mathcal{L}_{pic} + \varepsilon \mathcal{L}_{seg} + \mathcal{L}_{RMSE}$$

The terms

- The PCOL, SCOLw, IT and RMSE terms are existing (prototypes, sample contrastive, class alignment, regression).
- **delta L_pic — NEW: per-image concept supremacy (fixes Moderate).**
- **epsilon L_seg — NEW: auxiliary segmentation (sharpens BUSI heatmaps).**

What to watch

- Start delta and epsilon SMALL (0.05, 0.5); DR training already oscillates with four losses.
- Success = Moderate top-1 rises off 0.31 (chance 0.27) toward 0.45+ WHILE accuracy holds (~0.61 DR).
- *Bring losses online one at a time so instability is diagnosable.*

Architecture change & roadmap

BioMedCLIP backbone removes the random projection; fixes sequenced for one retrain

BioMedCLIP vision backbone

- Replace EfficientNet-V2S with BioMedCLIP's ViT vision tower.
- Image and text then share one pretrained medical space — the 512→1280 random projection disappears.
- *Does NOT fix Moderate alone — that still needs L_pic.*

Do it in ONE retrain

- BioMedCLIP backbone + L_pic + (BUSI) L_seg together.
- One training cycle instead of three; also reveals how the fixes interact.
- *Caution: BUSI is small (780); a ViT may train worse there — watch BUSI accuracy.*

Status of the fixes

- DONE (post-hoc, verified): push-apart (WHY), spatial concept heatmaps (WHERE), seg-guided localization.
- CODED, NOT YET RUN: per-image contrastive loss — no Moderate result until the retrain runs.
- OPEN: auxiliary seg loss, BioMedCLIP backbone, IDRiD localization eval.

So far, we've judged ESPAOCL using ground-truth labels we own.

Five LLM protocols on Claude Opus 4.7 — four blind, one labeled.

Independent LLM audit: does the explanation hold up?

An outside model judges coherence only — never diagnoses — and corroborates the localization fix

Model explanation
top concept + heatmap



LLM auditor
coherence only, no diagnosis



Coherence verdict
concept ↔ location consistent?

Raw heatmaps

38%

auditor accepts the concept, rejects the diffuse location

Segmentation-guided

88%

focused heatmap → location now matches the named feature

Three independent methods converge on the same finding:

energy-in-lesion 0.29 → 0.69 · visual panels diffuse → tight · LLM coherence 38% → 88%. Concepts were always correct; raw localization was the weak link; segmentation guidance fixes it. (n=8 BUSI sample; guided uses the predicted mask)

Future work, in priority order

Three method fixes, one validation step — each addresses one specific limitation we found

1

Tighten heatmaps

Auxiliary segmentation head OR weighted layer fusion → addresses diffuseness complaint

2

Per-image margin loss

Replace per-class prototype alignment with per-image contrastive → addresses contradictory ranking

3

BioMedCLIP vision tower

Eliminates the 512→1280 random projection, brings medical inductive biases

4

Re-run protocol 5

If complaints disappear, expect coherence on correct cases to rise from 30% toward 70-80%

5

Clinician reader study

Two ophthalmologists or radiologists, ~30 cases, same labeled outputs → confirm or refine LLM result

What is ours, stated precisely

We build an explainability layer on an inherited grading model; we do not claim the grader

Inherited (the base framework)

- Hybrid contrastive ordinal grading model: PCOL + SCOLw losses, regression head.
- EfficientNet-V2S backbone, 10-fold DR / 5-fold BUSI protocol.
- The grading accuracy under class imbalance.
- *It produces a grade only — no concepts, no heatmaps, no explanation.*

Ours (the explainability layer)

- The WHY+WHERE framework: clinical concepts (BI-RADS / ICDR) scored against the spatial features, with a per-concept heatmap each.
- Per-image contrastive concept loss with an ordinal margin (targets the Moderate grade).
- Auxiliary segmentation loss to sharpen attention at its source.
- Three-way localization validation, including the independent LLM coherence audit.

On the attribution tools specifically:

Grad-CAM++, layer fusion, and vision-language concept scoring are existing methods. The novelty is the framework that combines them to make this opaque ordinal grader explain itself in clinical terms, plus the two training losses — not a new attribution algorithm.

Takeaways & Future Work

What This Work Establishes

Interpretability gains are real

Concept scores >0.93 on all class-defining BUSI features; DR margin $+0.117$ ($p < 10^{-5}$)

Accuracy-explainability trade-off is measurable

-11.9 pp on DR; consistent across all 10 folds

Current setup is not jointly optimized

Semantic and regression objectives conflict under imbalance

Future Directions

Replace backbone

Use BioMedCLIP's vision encoder to eliminate the random 512→1280 projection entirely

Domain adaptation

Selective unfreezing of last 2–3 BioMedCLIP transformer blocks on DR

Joint optimization

Align semantic and regression objectives to enable higher effective γ without accuracy collapse